Article Review Assignment

Owen Malone

6.S058 Intro to Computer Vision

Dr. Rowan

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Research Context

The peer reviewed article I am choosing to review is Visually Indicated Sounds by Owens et al. [3]. The problem of generating audio from silent video is framed by Owens et al. as a problem of physical inference. That is, given only the visual appearance of an object and the action being performed upon it, can a model learn what sound that interaction should produce? This framing is significant because it positions the task not merely as audio generation, but as material recognition stemming from visual features. Rather than labeling objects by their material properties, the model discovers these properties on its own by learning trends between visual and auditory signals. This is an interesting research direction because it mirrors the way humans instinctively associate the visual texture and structure of an object with the sounds it produces. This is particularly useful for the field of computer vision because it demonstrates that visual features alone can provide enough information about physical material properties to serve as a signal for audio prediction. My own project similarly takes silent video of object interactions as input and attempts to predict the resulting sound. This setup provides both a theoretical motivation and a practical starting point for my project.

The paper attributes its design decisions to and draws inspiration from how infants learn and associate motion with sounds. Thus, Owens et al. builds upon works from Baillargeon [1] and Schulz [4], but expands upon their ideas and introduces them to the computer vision world. The dataset generated by Owens et al. for this project, called *Greatest Hits*, consists of videos and corresponding audios of different materials being hit by a drumstick. This experimentation process interestingly mimics how infants learn. The paper also sets itself apart from other methods used to blend the audio and visual domains. The paper cites multiple other projects that estimate material properties through vibrations and even a method of training a neural network on both modalities [2], but claims to use much different infrastructure to achieve true audio generation. However, the paper interestingly omits related work on audio generation methods, and tends to focus on classification, recognition, and representation learning.

One assumption that the paper makes in its setup is worth examining. By restricting interactions to a single drumstick hitting objects in an isolated system, the paper trades generalization for experimental control. Audio-visual correspondences in real life are much more complex. Sounds can come from partial occlusions, complex multi-object interactions, or even off-screen origins. The resulting model learns to predict impact sounds well under these constrained conditions, but generalization to unconstrained video remains an open question that I wish to tackle in my project.

Methodology and Technical Approach

The most prevalent technical choices Owens et al. made in the paper were: choice of dataset, audio representation, and machine learning methodology. The aforementioned *Greatest Hits* dataset is a great choice to study material properties and to train a model to generate sound in an isolated system. However, this does not scale well to real-world sound, which does not always come from drumsticks. For my project, I think it could be useful to train a model on this dataset to observe the kinds of sounds that are made by physically unique objects. The paper's choice of audio representation, namely cochleagrams, could also prove useful in my own project. A cochleagram is a time-frequency representation of audio that is modeled after the human inner ear (cochlea). Unlike a standard spectrogram, which uses linearly spaced frequency bins, a cochleagram uses a filter bank that mimics the frequency sensitivity of the human ear, spacing frequencies logarithmically and emphasizing lower frequencies where human perception is most sensitive. This makes cochleagrams a more perceptually meaningful representation of sound, and a natural choice for a task where the goal is to generate audio that sounds realistic to a human listener. For my project, which aims to predict sounds based on visual information, a representation like the cochleagram may be worth exploring, as the goal is ultimately to produce audio that is convincing to a human listener rather than accurate by some numerical metric.

To predict audio features from video, Owens et al. uses a recurrent neural network, specifically an LSTM, to map visual features extracted by a CNN to cochleagram features over time. The choice of an LSTM is well motivated here because impact sounds are temporal. Just as the initial sound is a characteristic of the material, the way sound decays also depends on this material. An LSTM is well suited to modeling these kinds of sequential dependencies. The CNN extracts spatial features from individual frames, while the LSTM reasons about how those features evolve across time, allowing the model to produce audio predictions that are consistent with the visual input. While this approach is well motivated, the added complexity of an LSTM may not be necessary if the choice of audio representation already encodes sufficient temporal information. Using a cochleagram may allow me to use a much simpler machine learning architecture than an LSTM, and is worth exploring as I architect my model.

Evaluation Assessment

Owens et al. evaluate their model using both a psychophysical study and auditory metrics. The psychophysical study, conducted on Amazon Mechanical Turk, presents participants with two videos from the *Greatest Hits* dataset of the same scenario, one with real audio and one synthesized. Participants must do their best to identify the real sound. The synthetic sounds from the model fooled participants 40% of the time. The model performed the best with softer materials like mud or dirt, compared to more uniform sounds coming from materials like metal or wood. Although this result is convincing, there is room for doubt. As part of the data report, participants were fooled 46.9% of the time when presented with a real sound drawn from the training set and played back out of context. This indicates that participants could still reliably identify synthesized audio most of the time. In addition, since the study was conducted entirely within the *Greatest Hits* dataset, it is unclear how well this evaluation reflects performance on real-world video, which is the very setting my project aims to address.

Use of AI Statement

To produce this paper, I used Claude for formatting and bibliography creation.

References

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